

■ SignLens Admin Web Dashboard Subsystem

Vite React Administrative Management Panel Setup Guide

Welcome to the **SignLens Admin Web Dashboard Subsystem**. This web-based management portal is engineered for platform administrators, course managers, and system auditors. Built with **React**, **Vite**, **Tailwind CSS**, and **Lucide Icons**, it provides real-time monitoring of translation logs, platform analytics, user accounts, and Learning Management System (LMS) course content.

■ 1. Directory Structure Overview

```
SignLens/admin/
├── src/
│   ├── components/
│   │   └── layout/          # Topbar, Sidebar, ProtectedRoute components
│   │   └── ui/             # Card, Table, Input, Button UI elements
│   └── hooks/              # Custom React hooks (useDashboard, useUsers, useLessons, useTranslations)
├── pages/                  # Admin view pages
│   ├── dashboard/         # System statistics, charts & accuracy meters
│   ├── translations/      # Real-time translation activity log table
│   ├── users/             # User account management & roles
│   ├── lessons/           # Sign language course & lesson editor
│   ├── quizzes/           # Interactive quiz builder
│   └── Activity_logs/      # Audit trails & system activity logs
├── services/              # Axios API client (api.js) & backend endpoints
├── App.jsx                 # React router application root
├── index.html              # Web entry HTML
├── vite.config.js          # Vite build configuration
└── package.json            # Dependencies & scripts
```

■ 2. Step-by-Step Installation & Local Setup

Step 2.1: Prerequisites

Ensure **Node.js (v18.0.0 or higher)** is installed on your host computer.

Step 2.2: Install Dependencies

Navigate to the `admin/` directory in Command Prompt / Terminal and run:

```
cd admin
npm install
```

■ 3. Starting the Admin Dashboard Server

Start the Vite development server:

```
npm run dev
```

Verification & Access:

- Terminal will output: Local: `http://localhost:5173/`
 - Open your web browser (Google Chrome, Microsoft Edge, or Firefox) and navigate to: `http://localhost:5173`
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■ 4. Admin Dashboard Features Walkthrough

- **Dashboard Analytics** (`pages/dashboard/Dashboard.jsx`):
 - Live system metrics (Total Users, Total Translations Performed, Accuracy Rates).
 - Real-time Accuracy Meters (`AccuracyMeter.jsx`) and daily usage charts (`DashboardChart.jsx`).
 - **Translation History Logs** (`pages/translations/TranslationLogs.jsx`):
 - View every sign language translation performed across the mobile app, including timestamp, user ID, predicted text (e.g., "THANK YOU"), and model confidence score (e.g., 96.2%).
 - **User Management** (`pages/users/Users.jsx`):
 - Manage user profiles, view registered accounts, and toggle administrative permissions.
 - **LMS Course Manager** (`pages/lessons/Lessons.jsx` & `pages/quizzes/Quizzes.jsx`):
 - Create, edit, and update sign language video lessons, course categories, quiz questions, and award badges.
 - **Activity Audit Logs** (`pages/Activity_logs/ActivityLogs.jsx`):
 - Inspect system security audit trails, login events, and operational events.
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■ 5. Common Admin Issues & Troubleshooting

■ Problem 1: Port 5173 is in use

- **Cause:** Another instance of Vite is running.
- **Solution:** Vite will automatically switch to port 5174. Access the app at `http://localhost:5174`.

■ Problem 2: Dashboard displays "Failed to load statistics"

- **Cause:** The Node.js Backend Gateway (Port 8001) is not running.
- **Solution:** Start the backend server first by running `npm run dev` inside the `backend/` directory.